

Databricks Platform Security Docs

Question: What guardrails govern Databricks Genie?



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Overview

This document provides an assessment of the existing controls and architectural constraints governing the Databricks Genie product (inclusive of Genie One, Genie Spaces, Genie Agent mode) as they relate to model safety and the role of external guardrails. Genie operates within a tightly scoped environment where the combination of architectural constraints, the underlying model's built-in safety alignment, and the product's inherent operational boundaries provides a security posture that addresses the concerns typically associated with general-purpose LLM deployments. While some of the models underpinning Genie have guardrails from the model provider, those are secondary to the core design constraints for the system. The sections below detail the specific technical and procedural controls currently in place.

Underlying Model Safety

Databricks Genie is powered by current models like Anthropic's Claude Sonnet and similar from Azure OpenAI. These demonstrate among the strongest baseline resistance to adversarial attacks of any commercially available large language models. An older independent red-teaming report by Holistic AI found that Claude 3.7 Sonnet blocked all 37 jailbreak attempts in their standardized audit—a 100% resistance rate—while GPT-4.5 achieved 97% resistance. Overall, these figures represent a clear generational trend: each successive model generation has demonstrated measurably improved safety characteristics, as shown in the table below.

Model Safety Improvements Across Generations

Metric	Earlier Generation	Current Generation	Improvement
Harmful output rate (OpenAI)	6.48% (GPT-3.5)	0.73% (GPT-4)	82% reduction
GCG attack success rate	~80% (2023 models)	~5% (mid-2024 models)	~94% reduction
Crescendo attack (Claude)	21–33% (Sonnet v1)	4–9% (Sonnet v2+)	~75% reduction
Holistic AI red-team audit	Varied (earlier models)	0% success (Claude 3.7)	100% resistance

Sources: OpenAI GPT-4 System Card; HarmBench evaluation framework; Holistic AI Red-Teaming Audit (2025).

On the HarmBench evaluation framework, Claude 3.5 Sonnet v2 achieved attack success rates of 0% against both GCG-transfer and AutoDAN attacks, and only 5.5% and 9% against the more sophisticated Crescendo and TAP methods, respectively. Claude Sonnet 4.5, the model powering Genie, represents the latest iteration in this trajectory of continuous safety improvement and is not a generic or early-generation LLM—it is a frontier model with state-of-the-art safety alignment built into its training process.

Architectural Constraints Eliminate Key Risk Vectors

The Genie product operates within a tightly scoped execution environment that neutralizes the primary risks that guardrails are designed to mitigate:

- **Read-only SQL execution:** Genie generates SQL queries that are executed in read-only mode. It is architecturally impossible for the application to perform INSERT, UPDATE, DELETE, DROP, or any other data-modifying operation. This eliminates the risk of data corruption, data destruction, or unauthorized data manipulation.
- **User-scoped permissions:** All generated queries execute within the context of the requesting user's existing permissions. Genie cannot access data that the user does not already have authorization to query. There is no privilege escalation vector—the application adds no incremental access beyond what the user already possesses through their standard SQL tooling.
- **No outbound connectivity:** The SQL execution environment operates with outbound network connections blocked. Even in a hypothetical scenario where a query could be crafted to exfiltrate data, there is no network path for that data to leave the environment. Data exfiltration via the Genie application is not technically possible.

Taken together, these constraints mean that the worst-case outcome of any interaction with Genie is that a user sees data they already have access to, formatted as query results. No data can be moved, copied externally, modified, or deleted.

Note on outbound connectivity: As of early 2026, a preview feature can be enabled to allow a Python UDF (not standard SQL) to perform outbound network operations. If enabled, allowed destinations are controlled via Serverless Egress Control for serverless DBSQL, or your own firewall rules for classic DBSQL.

Implicit Application-Level Guardrails

Genie's design incorporates guardrails that are more context-aware and effective than generic external solutions. While Genie's primary function is SQL query generation, it can also produce text responses—such as summarizing results, highlighting specific elements from query output, or providing analytical commentary. This text generation is scoped to the data context of the Genie space and is not a general-purpose knowledge or question-answering capability. The following controls apply across both SQL and text outputs:

- **Topic relevance validation:** As part of the system prompt, Genie validates whether a user's question is relevant to the tables, schemas, and metadata configured in the Genie space. Questions unrelated to the available datasets are rejected outright. In a live demonstration with a financial services organization, a user asked a Genie space “who is going to win the Premier League,” and the model responded that the question was unrelated to the available tables and asked for a relevant query instead. This behavior provides inherent protection against misuse—requests for harmful content (e.g., instructions for building weapons, generating offensive material) are rejected not because a guardrail flagged them as unsafe, but because they are fundamentally unrelated to the data the space is configured to serve.
- **Data-scoped output:** The model is designed to generate SQL queries and accompanying text responses grounded in the configured datasets. If a user attempts to elicit responses outside the scope of the Genie space's data context—for instance, asking the model to produce arbitrary

content unrelated to the available tables—the system rejects the request and directs the user back to relevant queries.

- **Hallucination resistance:** The topic validation mechanism also serves as a strong hallucination control. Because the model is constrained to generate queries and responses grounded in known schemas, this significantly limits the risk of the model fabricating information about topics outside the scope of the configured datasets. This is a more robust form of grounding than external guardrails typically provide.

These implicit guardrails are superior to standardized external solutions because they are deeply integrated with the application's purpose and data context. A generic guardrail system would have no awareness of which tables are configured in a Genie space or whether a query is relevant to the available data—it would only be able to apply broad content filters that are both less precise and more likely to create false positives.

Residual Risk Is Minimal and Mitigated

No system is immune to adversarial attack. It is theoretically possible that a sophisticated user could employ advanced prompt engineering techniques to bypass the topic relevance validation or output scoping constraints. Before addressing the likelihood and impact of such bypasses, it is worth examining what novel data a user could potentially access through Genie that they would not otherwise have visibility into.

Novel Data Exposure Analysis

Because Genie operates within the user's existing data permissions, the only information a user could potentially access that they would not already have through standard SQL tooling falls into two narrow categories:

- **Genie space instructions:** Genie space administrators can configure instructions that guide the model's behavior within a space—for example, specifying which datasets to prioritize, what types of questions to answer, or how to interpret certain fields. There is a marginal risk that a user could prompt the model to reveal these instructions. However, the practical impact of such exposure is extremely low. These instructions are inherently operational context relevant to the data and analysis the user is already authorized to perform. Exposure of these instructions does not constitute a meaningful security risk.
- **Databricks system prompt:** As with any AI-powered application, there is an inherent risk that a user could discover the underlying system prompt through prompt engineering techniques. This is a known and accepted characteristic of LLM-based systems that cannot be fully eliminated with or without external guardrails—guardrail products do not solve this problem. Databricks is not concerned about this risk because the value of the Genie product does not reside in the system prompt or in the model itself. The product's value is in how customers' own data context is provided to the model, how queries are generated and executed against Databricks SQL clusters, and how results are returned to enable the user to work effectively. System prompt visibility does not compromise the security of either Databricks or its customers.

Bypass Likelihood and Impact

Even setting aside the limited data exposure, the residual risk from any behavioral bypass is negligible for several reasons:

- **No incremental capability:** Even if a user successfully prompted Genie to produce content outside its intended scope, this would not grant them any capability they do not already have. Any information obtainable through a Genie bypass is trivially available through direct use of ChatGPT, Claude, or any other publicly accessible LLM. The Genie application adds no unique risk surface for this class of misuse.
- **Intentional effort required:** Subverting the multiple layers of protection in Genie would require deliberate, sophisticated prompt engineering. This is not a scenario where a casual user accidentally triggers harmful output—it would require someone actively and knowingly attempting to misuse the system.
- **Internal-only exposure:** Genie spaces are available only to authenticated internal users within the organization, or via the Genie Conversations API managed through a customer-provided application. This is not a public-facing, open-access system. The user population is known, credentialed, and subject to organizational policies and employment agreements. This dramatically reduces the likelihood and motivation for adversarial misuse.
- **Auditability and monitoring:** Genie space administrators have visibility into all messages sent by users within their spaces, providing a built-in audit trail. Any attempt to misuse the system can be traced back to a specific authenticated user. Additionally, many consumers access Genie spaces via the Conversations API, which provides a programmatic integration point where customers can implement their own additional input/output filtering if desired.

Why External Guardrails Are Not Used

Generic model guardrail solutions are designed for broad, general-purpose LLM deployments where the model has wide latitude to generate arbitrary content and where the output is consumed directly by end users or the public. Genie fits none of these criteria. Its primary output is SQL queries and data-grounded text responses. Its execution environment is read-only and permission-scoped. Its user base is internal and auditable. Its topic scope is constrained to configured datasets. The implicit guardrails embedded in Genie's design—topic relevance validation, data-scoped output, permission-scoped execution, and network isolation—are more targeted, more context-aware, and more effective than what any external guardrail product can offer for this specific use case.

Beyond the limited applicability of generic guardrails to this architecture, there is a practical constraint that prevents their adoption. Databricks system is architected such that any guardrail configuration must apply uniformly across all customers. This creates a direct conflict with the diverse requirements of the customer base. For example, a content filter that blocks discussions of self-harm would prevent organizations like Crisis Text Line from using Genie to support their suicide prevention services. Filters that restrict conversations about gambling would impact customers operating in the regulated gaming industry. Similarly, defense and military customers require the ability to discuss munitions, ordnance, and related topics as part of their legitimate analytical workloads. A one-size-fits-all guardrail configuration

would inevitably degrade product functionality for some segments of the customer base, without providing meaningful incremental security given the controls already in place.

More information

For more information about Databricks security, please see our [Security and Trust Center](#). The [Security and Trust Overview White Paper](#) will provide you a quick glimpse of our security controls, and you can find more detail within our [product documentation](#) and in our Enterprise Security Guide, along with compliance reports like our SOC 2 Type II. Your Databricks representatives will be happy to assist you with any resources you need.

Revision history

Date	Author	Description of changes	Version #
2026 Feb 05	DV	Initial version	1.0
2026 Jun 16	DV	Updated for newer model choices	1.1